

Sheik Kather Fahima

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Work Experience

UNIVERSITY OF APPLIED SCIENCES IN UPPER AUSTRIA, FH HAGENBERG, AUSTRIA

Bachelor's Research & Development Intern

June 2025 - July 2025

- Conducted in-depth research on **De-Nisting in Digital Forensics**, documenting the complete workflow from file collection to hash comparison.
- Evaluated multiple open-source tools for forensic imaging, hashing, and hashset matching, highlighting their features, performance, and forensic reliability.
- Investigated the role of the **National Software Reference Library (NSRL)** and its RDS dataset in supporting de-nisting practices.
- Explored **parallelization strategies** for hashing and large-scale hash set matching to improve efficiency in high-performance and multi-core environments, while identifying risks and limitations in applying parallelization to forensic imaging.

PWC, LINZ, UPPER AUSTRIA

Bachelor Intern (In collaboration with FH OÖe Hagenberg, University of Applied Sciences in Upper Austria)

July 2025 – August 2025

- Collaborated in a research team to initiate the development of an **AI-based law compliance reporting software**, leveraging the European Union law library to access and integrate regulatory information.
- Contributed to the system and report design, structuring the compliance report format to align with regulatory requirements, and assisted in shaping the application/website layout to ensure clarity, usability, and consistency in presenting compliance data.

Education

DE MONTFORT UNIVERSITY, DUBAI

Dubai Internet City, Dubai

- Course: BSc. (Hons) Cyber Security
- Graduated with a First-class Degree with Honours.
- Relevant Coursework:
Programming, Computer Networks, Web Application development, Database design and

Implementation, Computer Architecture, Incident/ Threat recognition and response, Cryptography, Penetration testing, Mathematics for Computing.

GULF ASIAN ENGLISH SCHOOL, SHARJAH

Muweilah, Sharjah

- Graduated with an 85% grade overall in the CBSE board examinations (2022)
- Subject Topper in Marketing

Major Projects

DE MONTFORT UNIVERSITY, DUBAI

Host-Based Penetration Testing

- Conducted a comprehensive **black-box penetration test** on a virtual machine provided by the university, following the NIST 800-115 methodology as part of the Penetration Testing module.
- Leveraged industry-standard tools, including Nmap for network discovery, Nessus for vulnerability scanning, and Metasploit for exploiting identified vulnerabilities.
- Identified and exploited critical vulnerabilities, such as weak passwords, insecure SSH configurations, NFS misconfigurations, FTP weaknesses, Rpcbind issues, and SMB vulnerabilities.
- Executed advanced attack techniques, including brute force attacks, reverse shells, and DoS/DDoS attacks, demonstrating the severity of identified security flaws.
- Provided detailed remediation strategies for each vulnerability, emphasizing:
 - The necessity of regular system updates and patch management.
 - Implementing strong password policies and secure system configurations.
 - Enforcing proper access controls to mitigate future risks.

Malware Detection Using Explainable AI (XAI) (Final year research Project)

- Undertook a project exploring advanced methodologies for malware detection with a focus on integrating Explainable Artificial Intelligence (XAI).
- Conducted a thorough literature review encompassing state-of-the-art malware detection techniques and XAI methods, aiming to bridge the gap between effective detection and interpretability.
- Designed a robust framework balancing high detection accuracy with transparency and interpretability in machine learning-based malware analysis.
- Implemented and evaluated models using Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, and SHAP/LIME for explainability.
- Evaluated detection and XAI techniques on real-world malware datasets, providing actionable insights and recommendations for optimal strategies in cybersecurity.
- Enhanced decision-making in cybersecurity by making model predictions interpretable, trustworthy, and actionable for professionals.

Student Leadership Activities

DE MONTFORT UNIVERSITY, DUBAI

Dubai Internet City, Dubai

Student Vice President of Education

2024 – 2025

Managing a team of class representatives, overseeing their selection and performance to ensure effective representation of student interests.

- Meeting with the university head monthly to discuss student issues and program feedback.
- Advocating for student needs in university decision-making.
- Acting as the primary point of contact for class representatives, addressing any issues or concerns they raise and providing guidance and support as needed.

Student Representative (Cyber Security)

2022 – 2024

- Acted as a liaison between students and faculty, addressing concerns and facilitating communication.
- Helped Students with academics and personal affairs.

DE MONTFORT UNIVERSITY, DUBAI

Dubai Internet City, Dubai

February 2024

Hacking the Future: Unravelling the bachelor's in cyber security, Presenter

- Held a Presentation to provide guidance to IYZ (International Year Zero) students for choosing a bachelor's degree in cyber security.

Referees

- Dr. Atheer Al-Mousa, Program Leader and Senior Lecturer (Cyber Security), Dept. of Cybersecurity, De Montfort University Dubai, atheer.al-mousa@dmu.ac.uk
- Professor Robert Kolmhofer, Professor and Head of Department Secure Information Systems, FH OÖe Hagenberg, University of Applied Sciences in Upper Austria, robert.kolmhofer@fh-hagenberg.at
- Professor Thomas Grurl, Professor at University of Applied Sciences Upper Austria, FH OÖe Hagenberg, University of Applied Sciences in Upper Austria, thomas.grurl@fh-hagenberg.at
- Christoph Pils, Senior Associate Regulatory & OT Security, PWC, Linz, Upper Austria, christoph.pils@pwc.com
- Roland Pucher, Senior Manager, Cybersecurity & Privacy, PWC, Linz, Upper Austria, roland.pucher@pwc.com